

The Proximity Prize Negative-Endpoint Ledger: Per-Field Assessments, Rescue Paths, and Sub- Lemma Close-Outs

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Proximity Prize — research ledger / companion note. **Date:** 2026-06-10. **Convention:** ABF δ -radius throughout (δ = relative proximity *radius*; $\rho = k/n$; Johnson radius $J = 1 - \sqrt{\rho}$; Singleton/MDS capacity $1 - \rho$; list-decoding capacity radius $R_{\text{cap}}(q, \rho) := H_q^{-1}(1 - \rho)$ — see §1).

0. What this ledger is

This companion note records **research-state assessments at mixed maturity levels; the status tags are load-bearing**. It is *not* a polished paper: rows below range from theorem-grade citations of the literature, through ASSEMBLED lemmas (cited components plus exact threshold arithmetic), through ESTABLISHED-MODULO-FORMALIZATION computational verdicts, down to INFERRED arithmetic flagged VERIFY and frankly OPEN items. Every claim carries its tag; a claim must never be quoted above its tag.

The polished companion is the short paper **technical-note-short.md** (same repository): the two grand challenges and their non-equivalence, the budget trichotomy in $B = 2^{-128}|F|$, the proven list-size bracket $[J - o(1), R_{\text{cap}}]$ for $|F| > 2^{128}$, the BCHKS Thm 1.9 separation, the field-agnosticity of the negative mechanism, and the open problems (keystone P'). Definitions (CA/MCA, interleaved codes, lists, the relation $(\star)$: $|\Lambda(C, \delta)| \leq |\Lambda(C^{\equiv m}, \delta)| \leq |\Lambda(C, \delta)|^m$) are as in the short paper §2. This ledger holds everything the short paper points to: the assembled per-field negative-ceiling **assessment** (internal revision rounds R13–R18), the 31/64/128/256-bit arithmetic and tables, the M31/BabyBear/circle-domain story, the 256-bit rescue-path assessments, the N1/N2 close-outs with their caveats, the experimental data, and the P' route summary.

Status-tag glossary. **PROVEN** — a theorem in the cited literature, used inside its stated range. **PROVEN BRACKET** — composition of PROVEN endpoints. **ASSEMBLED** — assembled from cited components plus exact threshold arithmetic; named soft spots listed. **ESTABLISHED-MODULO-FORMALIZATION (EMF)** — exact computation plus a structural argument, not yet written as a theorem; fail-closed. **CONDITIONALLY ESTABLISHED** — proven modulo one named, explicitly stated extension of a cited lemma. **OUR-DERIVED** — our composition/arithmetic on cited inputs. **INFERRED (VERIFY)** — arithmetic inferred from a construction's parameter window; explicitly *not* a proven endpoint. **EXPERIMENTAL** — exact small-field enumeration, corroborative only. **CONJECTURED / OPEN** — what they say.

1. Conventions

The ceiling convention (R_{cap} , load-bearing). Two nearby radii appear in the literature and in our tooling: the **exact inverse-entropy crossing** $R_{\text{cap}}(q, \rho) := H_q^{-1}(1 - \rho)$ — the radius δ solving $H_q(\delta) = 1 - \rho$, the convention of ledger §3b / `calculator/proximity_parameters.py` — and the **large- q approximation** $1 - H_q(\rho)$. **Every proven upper-ceiling statement in this ledger is a statement about R_{cap}** ; the large- q approximation appears only where explicitly labeled as such. The two are **not equal at any deployed rate** (verified by direct computation): the gap is ≤ 0.0017 across deployed rates and field sizes (maximum ≈ 0.00166 at 31-bit, $\rho = \frac{1}{8}$), with rate-dependent sign — the approximation sits *below* R_{cap} at $\rho = \frac{1}{2}$ (by $+9.6 \cdot 10^{-5}$ at 31-bit, $+1.7 \cdot 10^{-7}$ at 256-bit) and *above* it at $\rho \leq \frac{1}{4}$ (by $\approx 2.0\text{--}2.3 \cdot 10^{-5}$ at 256-bit) — i.e. the two differ **beyond the fifth decimal** at some rates even at 256-bit, and the approximation can overstate the proven ceiling, which is exactly why R_{cap} is the proven object. The large- q approximation sits a distance $H_q(\rho) - \rho \leq 1/\log_2 q$ below Singleton capacity (CS25 Claim 1; $\leq 1/31$ for 31-bit fields).

A historical software caveat — now FIXED (R29/M2 follow-up). The resolver `calculator/listsize_resolution.py` formerly capped the alphabet at `_QCAP_LOG2 = 200` before evaluating H_q , so its generated “256-bit” ceiling entries were in fact **200-bit** values — e.g. 0.49500 at $\rho = \frac{1}{2}$ vs the true 256-bit large- q -approximation value 0.49609. The cap is now 320 (no deployed field is capped) and the resolver reports the proven ceiling at the field’s true q ; all 256-bit ceiling values quoted in this ledger are true $q = 2^{256}$ values, matching `problem-ledger.md` §3b and the regenerated `listsize-analysis.md` §8. The exact 256-bit crossings R_{cap} are 0.49609/0.74681/0.87285/0.93616 at $\rho = \frac{1}{2}/\frac{1}{4}/\frac{1}{8}/\frac{1}{16}$ (the large- q approximation would give 0.49609/0.74683/0.87288/0.93618).

The notation clash (carry the ABF convention). ABF/CS25/GG25 use δ for the proximity *radius*; BCHKS and CGHLL use δ for the **code’s minimum distance** ($= 1 - \rho$, MDS) and γ (resp. θ) for the radius. BCHKS’s soundness error is a/q where $a = \#\{z : \Delta(f + zg, C) \leq \gamma\}$ and $q = |F|$; its “proximity loss” is ε^* (the gap between hypothesis and conclusion radii). When importing BCHKS into ABF: radius $\delta \equiv \gamma$, rate $\rho \equiv 1 - \delta_{\text{BCHKS}}$, error a/q with a a count. **CS25’s “capacity” is the list-decoding capacity** (our R_{cap} , which they state via the large- q approximation), strictly below the Singleton $1 - \rho$ that ABF/BCHKS/GG call capacity. **CGHLL’s “Elias radius” is likewise the large- q approximation $1 - H_q(\rho)$** ; wherever we quote their conjectures we write R_{cap} , the difference (≤ 0.0017) being below the resolution of every claim involved. All statements below are in the ABF convention.

2. The MCA bracket in detail (sub-problem 1)

This section records the *bracket* $[\delta_{\text{known-positive}}, \delta_{\text{known-negative}}]$ for the MCA challenge; the true δ_C^* lies inside it, and pinning it is open. **This is a bracket, not a theorem-complete determination.** Scoping rule (carried from the short paper §5, load-bearing): the known bracket *endpoints* are field-type insensitive; the negative *mechanism* is field-agnostic (§3.2); the *positive* beyond-Johnson program is **not** field-agnostic (§3.5).

2.1 Positive side — up to the Johnson radius (all fields)

[PROVEN] (Bordage–Chiesa–Guan–Manzur, eprint 2025/2051, Thm. 9.2). For $C = \text{RS}[F, D, k]$ over *any* domain D (including smooth subgroups), the affine-line / univariate-powers generator, message degree $d = k - 1$, and integer $m \geq 3$:

$$\varepsilon_{\text{mca}}(C, \delta) \leq \frac{(m + \frac{1}{2})^7 n^2 d}{3 \rho^{3/2} |F|}, \quad \text{valid for } \delta \leq 1 - (1 + \frac{1}{2m})\sqrt{\rho}.$$

Loss-free (= ABF ε_{mca}); as $m \rightarrow \infty$ the validity window $\rightarrow J = 1 - \sqrt{\rho}$, but larger m inflates the error via $(m + \frac{1}{2})^7$ (the m -vs- δ trade).

[PROVEN] (BCHKS, eprint 2025/2055, Thm. 1.5 = ABF Thm. 4.12 core). With $\eta = 1 - \sqrt{\rho} - \delta > 0$:

$$\varepsilon_{\text{mca}}(C, \delta) \leq \frac{C_\rho n}{\eta^5 |F|}, \quad \text{valid for } \delta \leq 1 - \sqrt{\rho} - \eta, \quad \varepsilon^* = 0.$$

The η -exponent 5 is firm (BCHKS Table 1); the leading constant C_ρ is an unspecified $O_\rho(\cdot)$ — [OUR-DERIVED caveat] we treat it as a VERIFY placeholder ($C_\rho = 1$ in the calculator). Linear in n (vs $n^2 d \sim n^3$ for Bordage–Chiesa), hence asymptotically stronger, but kept unverified until C_ρ is pinned.

Field-size consequence [OUR-DERIVED]. The *verified* Bordage–Chiesa numerator $n^2 d$ costs $\approx 2 \log_2 n + \log_2(\rho n)$ bits (≈ 60 at $n = 2^{20}$, ≈ 90 at 2^{30}). Certifying $\varepsilon_{\text{mca}} \leq 2^{-128}$ from this single-code bound therefore needs $\log_2 |F| \gtrsim 256$ for n up to 2^{30} : M31/BabyBear/Goldilocks **cannot** certify 2^{-128} from the single-code MCA bound at all (see `calculator/out/delta_star_tables.md`). Those deployments reach 128-bit soundness via the query term $(1 - \delta)^t$ plus repetition/PoW and by running the soundness argument in a large extension — not from a single-code 2^{-128} MCA certificate. (BCHKS Thm 1.5’s linear- n bound would relax this *if* C_ρ were pinned.)

$\Rightarrow \delta_{\text{known-positive}} = J = 1 - \sqrt{\rho}$ for smooth RS over all fields — **as a radius-level theorem** ($\varepsilon_{\text{mca}} \leq \text{poly}(n) d / |F|$ up to J). Whether that error meets the prize target $\varepsilon^* = 2^{-128}$ is a *separate, field-size* question: per the consequence above, only ≥ 256 -bit fields certify it from the single-code bound. Throughout this ledger, “positive to Johnson” at 31/64/128-bit fields means the radius-level theorem, **not** a certified $\varepsilon^* = 2^{-128}$ lower bound.

2.2 Negative side — near capacity (the mechanism is field-agnostic: prime and extension)

[PROVEN] (Crites–Stewart, eprint 2025/2046, Cor. 1 = ABF Thm. 4.17). For $q \geq 10$, $q \geq n$, in a k -window reaching down to the list-decoding capacity, there exist $u^{(0)}, u^{(1)}$ with $\Delta(u^{(1)}, C) > \delta$ yet $\Delta(u^{(0)} + \lambda u^{(1)}, C) \leq \delta$ for every $\lambda \in F$. Thus $\varepsilon_{\text{ca}} = 1$ reaching down to the list-decoding capacity radius R_{cap} ; δ_C^* is provably strictly below Singleton capacity.

[PROVEN] (Kambiré, arXiv:2604.09724, Thm. 1; = the KK25 / CGHLL Thm. 37 construction). For every $C > 0$ and rate $\rho \in (0, \frac{1}{2})$ there are infinitely many n, k with $C = \text{RS}[F_p, \langle \omega \rangle, k]$, $\langle \omega \rangle$ a smooth subgroup of order $n = 2^t$, prime $p \equiv 1 \pmod n$, $p < n^A$, and words $f = X^{rm}$, $g = X^{(r-1)m}$ such that at

$$\delta = (1 - \rho) - \frac{2}{K \log n} \quad (\eta = 2/s = 2/(K \log n) \text{ below capacity})$$

there are $\geq n^C$ scalars z with $\Delta(f + zg, C) \leq \delta$, yet $\Delta([f, g], C^2) > \delta$ — a direct proximity-gap and CA failure. [OUR-DERIVED, INFERRED] **Finite-field caveat (key number) — a finite-field inferred ceiling, not a proven endpoint:** the construction needs a prime $p \in [4^s, 8^s]$, so $s \leq \log_2 |F|/3$ and the gap is a *constant* $\eta \geq 6/\log_2 |F|$ (≈ 0.047 at 128-bit, ≈ 0.023 at 256-bit), **not** $o(1)$ at deployed sizes. The calculator encodes this as $\delta_{\text{unsafe}} = (1 - \rho) - 6/\log_2 |F|$. **This continuum endpoint is inferred arithmetic from the $8^s \leq |F|$ window — flagged VERIFY — and is retained only as an explicitly-labeled *pre-R13 continuum* asymptotic reference (the R13 power-of-two mechanism-asymptotic reference at 256-bit is $(1 - \rho) - 2/64$, §2.4), not a proven δ_{unsafe} ; the proven points underneath it are the positive-to-Johnson floor (radius-level, §2.1) and the unconditional BCHKS M31 witness at $\delta = \frac{1}{2}$ (over the non-smooth domain F_q^* — smooth-subgroup landing VERIFY; below). The per-field threshold version is the assembled Lemma below (R13): $\delta_C^* \leq (1 - \rho) - 2/s_{\max}(b)$ with $s_{\max}(b)$ a power of two. (Kambiré states it for prime F_p with $\rho \in (0, \frac{1}{2})$ open; the *mechanism* is established field-agnostic at EMF grade — see §3.2 — and the $\rho = \frac{1}{2}$ boundary is covered by the unconditional M31 BCHKS witness.)**

Lemma (per-field instantiated negative ceiling — ASSEMBLED, R13). Let $F = F_p$ be a prime field with $b = \log_2 p$ bits, let the smooth evaluation domain have order $n = 2^t$ with $n \mid p - 1$, and let $s = n/m$ be the Kambiré quotient parameter — necessarily a **power of two** (R13 integrality). Suppose (i) **distinctness:** $p > \varphi(s)^{\varphi(s)}$ (the KK25 distinctness hypothesis; **cited, not reproved here**); (ii) **count:** $N(s, \rho) > 2^{b-128}$, where $N(s, \rho)$ is the per-rate bad-scalar count — Kambiré’s $N0_{\text{sum}}$ for $\rho < \frac{1}{2}$, and the proven antipodal count $3^{s/2}$ for $\rho = \frac{1}{2}$ (this case carries the **N1 conditionality flag**, §3.3); (iii) the construction’s standing distinctness/degree hypotheses; (iv) **radius/size:** s large enough that the

firing radius stays strictly above Johnson, $(1-\rho)-2/s > J = 1-\sqrt{\rho}$, i.e. $s > 2/(\sqrt{\rho}-\rho)$ — numerically $s \geq 16$ suffices for all four deployed rates (the threshold is ≤ 10.7 , attained at $\rho = \frac{1}{16}$) — together with Kambiré’s standing parameter hypotheses ($s = n/m$ with $s \mid n$, r in the construction’s admissible range). Without (iv), count + distinctness alone would nominally admit small- s “witnesses” at radii at or below Johnson, contradicting the proven positive side (§2.1); the lemma is asserted only in the near-capacity band above J . (All tabled $s_{\max}$ values, 16/16/32, satisfy (iv).) Then $\varepsilon_{\text{mca}}(C, \delta) > 2^{-128}$ at $\delta = (1-\rho) - 2/s$, and hence (using monotonicity of the failure event in δ , as Kambiré/CS25 state their ceilings)

$$\delta_C^* \leq (1-\rho) - \frac{2}{s_{\max}(b)},$$

where $s_{\max}(b)$ is the largest power of two satisfying (i)+(ii): $s_{\max} = 16$ at $b = 31$ — anchored at **BabyBear** ($p = 15 \cdot 2^{27} + 1$, 2-adicity 27), where the lemma applies verbatim; **literal M31 caveat:** $F_{2^{31}-1}$ has $p-1 = 2(2^{30}-1)$ with $v_2(p-1) = 1$, so it has **no smooth domain of order ≥ 4** and the lemma is vacuous over literal M31 — the M31 figure refers to the deployed circle/ F_{p^2} smooth subgroup of order $p+1 = 2^{31}$ via the §3.2 field-agnostic mechanism — so the M31 row is properly read as **31-bit characteristic / 62-bit alphabet** (F_{p^2}), not a literal 31-bit-field row (distinctness calibrated at the characteristic $p \approx 2^{31}$, giving the same $s_{\max} = 16$; count threshold 2^{b-128} trivial at the alphabet’s $b = 62$), that row inheriting N2’s **ESTABLISHED-MODULO-FORMALIZATION** grade rather than the prime-field ASSEMBLED grade — $s_{\max} = 16$ at $b = 64$, $s_{\max} = 32$ at $b = 128$. At $b = 256$ **no valid s exists** (distinctness allows $s = 64$ but the count $3^{32} = 2^{51} \ll 2^{128}$; for $\rho < \frac{1}{2}$ the $N0_{\text{sum}}$ count is smaller still — `n2_prize_threshold.json` shows all four rates infeasible), so the mechanism establishes **no threshold ceiling** there. *Status: ASSEMBLED from cited components* — Kambiré *arXiv:2604.09724 Thm. 1* (bad-scalar count) + KK25 distinctness lemma (cited, not re-proved) + exact threshold arithmetic (`experiments/small_rs_atlas/n2_prize_threshold.py`) + the R13 power-of-two integrality correction. *Soft spots:* the literal KK25 distinctness lemma is cited, not re-proved; the $\rho = \frac{1}{2}$ case is N1-conditional; the deg-convention ($N0_{\text{fire}}$ strict vs $N0_{\text{sum}}$ Kambiré-literal) is noted and tracked per row. (The continuum $(1-\rho) - 6/\log_2 |F|$ of §2.4 is **not** part of this lemma — it is retained only as an explicitly-labeled asymptotic reference; the bracket’s lower end J and the M31 $\delta = \frac{1}{2}$ point remain proven.)

[**PROVEN, conditional**] (**BCHKS Thm. 1.13**). Prime multiplicative-subgroup RS at $\gamma = \delta_{\min} - \Theta(1/\log n)$ forces proximity loss $\Theta(1/\log n)$ — conditional on additive-combinatorics Conjecture 1.12. [**PROVEN, uncon-**

ditional — over the *non-smooth* domain F_q^* ; **smooth-domain landing VERIFY**] for M31, $\rho \approx \frac{1}{2}$: $\text{RS}[F_{2^{31}-1}, F_q^*, (q-1)/2 + 2/\log_2 q]$ has f, g that are $\frac{1}{2}$ -close for *all* z yet CA-distance $\geq \frac{1}{2} + \frac{1}{62} \approx 0.516$ (loss ≈ 0.016) — a genuine failure at radius $\delta = \frac{1}{2}$ for a deployed prime at rate $\approx \frac{1}{2}$, **but not, as stated, a smooth-domain prize witness**: F_q^* has order $q-1 = 2(2^{30}-1)$, not a power of two, so it is not smooth in ABF’s sense, and whether the witness lands on a strictly power-of-two subgroup is **VERIFY** (per `bchks.md`). Wherever this ledger cites “the unconditional M31 witness”, it is this F_q^* (non-smooth-domain) statement.

$\Rightarrow \delta_{\text{known-negative}}$ (**prime**): per-field $(1-\rho) - 2/s_{\max}(b)$ per the assembled Lemma above (R13; KK25 cited not re-proved, $\rho = \frac{1}{2}$ N1-conditional); the continuum $(1-\rho) - 6/\log_2 |F|$ remains a *finite-field inferred ceiling* (inferred from the window arithmetic, VERIFY, asymptotic reference only; *not* a proven endpoint, though the M31 $\delta = \frac{1}{2}$ point below it is proven — over the non-smooth F_q^* domain, smooth landing VERIFY); the **same inferred ceiling holds for genuine odd-characteristic extension fields** F_{p^e} (the smooth-domain counterexample *mechanism* is field-agnostic — its distinct-bad-scalar count is a characteristic-zero cyclotomic invariant, verified exactly; see §3.2 and `n2-verdict.md`). Extensions buy no advantage; the negative forecloses the same constant- $\eta = 2/s$ near-capacity band for both.

2.3 The barrier: BCHKS Theorem 1.9

[**PROVEN**] (**BCHKS Thm. 1.9 = ABF Thm. 5.2; all RS**). For $\gamma = \text{LDR}_{F,D,q}(\delta_{\min}) + 2/n$ (just past the list-decoding radius *for list size* q), there exist f, g with $\geq q/(2n)$ close line points but $\Delta([f, g], C^2) \geq \delta_{\min} - 1/n$, so $a/q \geq 1/(2n)$ independent of q . **Consequence**: improving MCA / proximity gaps beyond the Johnson radius for *any* RS code **requires** first improving its list-decoding radius (at list size q) beyond Johnson — a separately hard, in-general-false-for-some-RS, long-open problem.

The barrier is *conditional, not absolute* (`line-decoding-analysis.md` §2.3): it does not forbid $\delta_C^* > J$; it proves that any *small-error* ($\varepsilon_{\text{mca}} \leq \text{poly}(n)/q$) certificate of $\delta_C^* > J$ for smooth RS is itself a list-decoding-beyond-Johnson result. The only crack is **large-error (M)CA** (error a/q not $\ll 1/n$ but still $\leq 2^{-128}$), which both CS25 and CGHLL flag as the realistic target. This is exactly why the prize poses *both* challenges. (The list-size challenge’s escape from this barrier is the short paper’s §4.)

2.4 The per-rate and per-field tables (R13–R18) and the 256-bit status

$\delta_{\text{known-positive}} = J$ is a field-type-insensitive endpoint and **proven**; the negative side’s *mechanism* is field-agnostic (the smooth-domain counterexample is a characteristic-zero cyclotomic invariant — §3.2, `n2-verdict.md`), so prime and genuine odd-char extension share one bracket — endpoint-wise; this says nothing about positive solution strategies (§3.5). **The negative endpoint**

δ_{neg} below is given by the assembled Lemma of §2.2 (R13) — $\delta_{\text{neg}} = (1 - \rho) - 2/s_{\text{max}}(b)$ with $s_{\text{max}}(b)$ the largest power-of-two s passing the KK25 distinctness calibration $p > \varphi(s)^{\varphi(s)}$ ($s_{\text{max}} = 32$ at $b = 128$); soft spots per the lemma’s status tag: KK25 cited not re-proved, $\rho = \frac{1}{2}$ N1-conditional. At $b = 256$ the mechanism is **infeasible at** $\varepsilon^* = 2^{-128}$ (count $3^{32} = 2^{51} \ll 2^{128}$), so there is **no threshold-established 256-bit ceiling below** R_{cap} (the generic CS25/Elias ceiling at $\approx R_{\text{cap}}$ stands) and the 256-bit mechanism-asymptotic reference is $(1 - \rho) - 2/64$ (R13 power-of-two convention, $s = 64$ the largest distinctness-allowed power of two — the primary reference convention used throughout this ledger, matching the N1 figures of §3.3; the pre-R13 continuum $(1 - \rho) - 6/256$ is retained only as a secondary continuum reference) — inferred, VERIFY — genuinely open, outside the assembled lemma’s reach; *not* a proven endpoint; the proven negative point is the unconditional M31 $\delta = \frac{1}{2}$ witness (non-smooth F_q^* domain, smooth landing VERIFY — §2.2). Values from `problem-ledger.md` §5 and `calculator/out/delta_star_tables.md` (target $\varepsilon^* = 2^{-128}$, $n = 2^{20}$; the generated δ_{unsafe} column is field-agnostic in mechanism — prime and extension rows show the identical inferred ceiling, no “open” upper end; note the generated column still shows the old continuum $(1 - \rho) - 6/\log_2 |F|$ values and awaits regeneration per R13).

Per-rate table (prime *and* genuine odd-char extension; the endpoints are field-type insensitive):

ρ	J (positive, proven, radius-level)	δ_{neg} @128-bit (assembled Lemma §2.2, R13: $s_{\text{max}} = 32$)	δ_{neg} @256-bit (mechanism-asymptotic only, NOT threshold-established — R13 power-of-two convention $(1 - \rho) - 2/64$, $s = 64$; pre-R13 continuum $(1 - \rho) - 6/256$ in brackets) gap @128-bit (prime <i>and</i> extension)	
1/2	0.29289	0.43750	(0.46875, reference only; continuum 0.47656)	[0.293, 0.438]

ρ	J (positive, proven, radius- level)	δ_{neg} @128-bit (assembled Lemma §2.2, R13: $s_{\text{max}} = 32$)	δ_{neg} @256-bit (mechanism- asymptotic only, NOT threshold- established — R13 power-of-two convention $(1-\rho)-2/64$, $s = 64$; pre-R13 continuum $(1-\rho)-6/256$ in brackets)	gap @128-bit (prime <i>and</i> extension)
1/4	0.50000	0.68750	(0.71875, reference only; continuum 0.72656)	[0.500, 0.688]
1/8	0.64645	0.81250	(0.84375, reference only; continuum 0.85156)	[0.646, 0.813]
1/16	0.75000	0.87500	(0.90625, reference only; continuum 0.91406)	[0.750, 0.875]

Per-field headline table at $\rho = \frac{1}{2}$ — split by domain type and evidence grade (R13). The two blocks are deliberately separate: prime-field smooth-subgroup rows carry the assembled Lemma’s ASSEMBLED grade; circle/extension *interpretation* rows carry N2’s ESTABLISHED-MODULO-FORMALIZATION grade and must never be quoted at ASSEMBLED grade.

(a) *Prime-field smooth-subgroup rows (assembled Lemma §2.2 applies verbatim — ASSEMBLED grade):*

field	b (bits)	$s_{\max}(b)$	ceiling $(1 - \rho) - 2/s_{\max}$ at $\rho = \frac{1}{2}$	grade
BabyBear ($p = 15 \cdot 2^{27} + 1$, 2-adicity 27)	31	16	0.375	ASSEMBLED (KK25 cited not re-proved; $\rho = \frac{1}{2}$ N1- conditional)
Goldilocks	64	16	0.375	ASSEMBLED (same soft spots)
128-bit prime	128	32	0.4375	ASSEMBLED (same soft spots)
256-bit	256	none valid	no threshold- established ceiling below R_{cap} (count $3^{32} = 2^{51} \ll 2^{128}$); mechanism- asymptotic reference 0.46875 only	OPEN below R_{cap} ; generic CS25/Elias ceiling at $\approx R_{\text{cap}}$ stands

(b) *Extension/circle-domain interpretation rows (N2 mechanism transfer — NOT theorem grade, NOT the assembled prime-field grade):*

field	b	$s_{\max}$	ceiling at $\rho = \frac{1}{2}$	grade
M31 (circle/ F_{p^2} reading; 31-bit characteristic / 62-bit alphabet)	31 (char) / 62 (alphabet)	16	0.375 — <i>via</i> <i>circle/F_{p^2}</i> <i>smooth</i> <i>subgroup of</i> <i>order</i> $p + 1 = 2^{31}$; $N2$ <i>ESTABLISHED-</i> <i>MODULO-</i> <i>FORMALIZATION</i> <i>grade, not</i> <i>theorem</i> <i>grade; literal</i> <i>M31 has</i> $v_2(p - 1) = 1$, <i>no smooth</i> <i>subgroup of</i> <i>order ≥ 4</i>	EMF (N2, §3.4; distinctness calibrated at the character- istic)

Radius-level caveat for all tables. The positive endpoint J is the radius theorem of §2.1; at 31/64/128-bit fields the single-code bound *cannot* certify $\varepsilon_{\text{mca}} \leq 2^{-128}$ at any radius (§2.1), so for those fields the table brackets the threshold radius δ_C^* — it is **not** a certified $\varepsilon^* = 2^{-128}$ two-sided bracket. Only at ≥ 256 -bit fields does the positive side carry a 2^{-128} certificate ($\delta_{\text{safe}} = 0.29251 \approx J$). At $\rho = \frac{1}{2}$, $n = 2^{20}$: 256-bit prime gives $\delta_{\text{safe}} = 0.29251 (\approx J)$; the query term alone needs $t = 257$ queries for 128 bits ($-\log_2(1 - \delta_{\text{safe}}) = 0.4992$, $128/0.4992 \approx 256.4$), while the calculator’s *composite* count is $t = 272$ — larger because at δ_{safe} the certified ε_{mca} already nearly saturates the 2^{-128} budget, so the query term must cover only the residual slack $2^{-128} - \varepsilon_{\text{mca}} - |\Lambda|/|F| \approx 2^{-135.8}$ ($272 \times 0.4992 \approx 135.8$ bits); a 256-bit extension gives the same δ_{safe} and the same δ_{unsafe} . M31/BabyBear/Goldilocks are **infeasible** for the single-code certificate (§2.1). The headline framing (64 $\rightarrow$ 128 bits at $\rho = \frac{1}{2}$; short paper §1) is precisely whether δ_C^* can be pushed above J inside these brackets.

Per-field corrections and the 256-bit status (internal revision rounds R13–R18; problem-ledger.md §5–§6 and the VERIFY register). This is the assembled per-field negative-ceiling assessment — R13 is the power-of-two integrality correction used throughout; R14–R18 are the 2026-06-10 rescue-path assessments. The R13 correction moves every previously tabulated per-field ceiling (values at $\rho = \frac{1}{2}$): at $b = 31$ the old continuum value 0.30645 was too *pessimistic* — the honest ceiling is $(1 - \rho) - 2/16 = 0.375$ ($s_{\max} = 16$ under the KK25 $\varphi(s)^{\varphi(s)}$ calibration; the direct $b = 31$ anchor is **BabyBear** — literal M31 has no smooth domain of order ≥ 4 , its 0.375 trans-

ferring via the deployed circle/ F_{p^2} smooth subgroup at N2’s ESTABLISHED-MODULO-FORMALIZATION grade, per the §2.2 Lemma’s M31 caveat — the M31 row is 31-bit *characteristic* / 62-bit *alphabet*; this re-scoping strengthens, not weakens, the withdrawal — the direction of the correction, window wider than previously claimed, is unchanged), so the M31 open window is $[0.293, 0.375]$ — a **radius-level** window: like every sub-256-bit field, M31 cannot certify $\varepsilon_{\text{mca}} \leq 2^{-128}$ from the single-code bound (§2.1), so this brackets the threshold radius, not a certified $\varepsilon^* = 2^{-128}$ quantity — and the earlier “**almost closed**” reading is **withdrawn**; at Goldilocks ($b = 64$) the old 0.40625 was too *optimistic* (corrected to 0.375, $s_{\text{max}} = 16$); at $b = 128$ the old 0.45313 falls to 0.4375 ($s_{\text{max}} = 32$).

The 256-bit rescue paths (R14–R18 assessments, ledger VERIFY register, 2026-06-10). At $b = 256$ the mechanism establishes **no threshold ceiling**, and the two rescue paths are assessed as follows. **(a) Per-prime distinctness certification — BLOCKED with current tools.** Full distinctness is not needed (only $\text{count}_p \geq 2^{128} = p^{1/2}$), but even under the most optimistic characteristic-zero count the required $s \approx 162$ means distinguishing $\sim \binom{81}{40} \approx 2^{77}$ subset sums by enumeration (computationally infeasible), and the height/norm route fails for a fixed prime ($|\text{Norm}(\Delta)| \leq (2s)^{\varphi(s)} \approx 2^{676} > 2^{256}$, so individual divisibility cannot be excluded, and no union bound over the $\sim 2^{154}$ difference pairs is available). **(b) BGK-type subgroup-sumset growth — OPEN, and not close** (literature/notes/subgroup-sumset-growth.md, surveyed 2026-06-10): the needed statement is that ~ 83 -fold subset sums of the order- s subgroup $\mu_s \subset F_p^*$ with $s \approx 324 = p^{0.033}$ cover $\geq p^{1/2} = 2^{128}$ values; the best published bounds give $\sim 2^{15}$ — an ≈ 113 -bit gap — with covering known only at $\sim 2^{63}$ summands (Cipra–Cochrane), and the $s \asymp \log p$ regime named open even for qualitative cancellation. (Scoping caveat: this negative verdict is specific to the *tiny-subgroup* regime of the 256-bit ceiling; the positive keystone P' lives at much larger subgroups, $|H| = n = p^{0.12} p^{0.65}$ deployed, and the verdict must not be over-propagated to it.) The computed feasibility boundary is $b^* \approx 140$ bits — a Kambiré-type prize-level witness exists only for $|F| \leq 2^{140}$ (experiments/small_rs_atlas/results/n2_prize_threshold.json; n2-verdict.md). The honest 256-bit status for a negative ceiling **below** R_{cap} (i.e., a Kambiré-type near-capacity, constant- η ceiling) is therefore **OPEN — needs a new idea**; the generic CS25/Elias ceiling at $\approx R_{\text{cap}}$ (§2.2) stands at every field size, 256-bit included; the per-field values above carry the assembled Lemma’s soft spots verbatim (KK25 cited not re-proved; $\rho = \frac{1}{2}$ N1-conditional; deg-convention tracked).

3. The line-decoding frontier: core-bet verdict and the sub-lemma close-outs

The positive MCA route relies on **line-decodability** (ABF Def. 4.20 / GG25 Def. 3.1): C is (δ, a, b) -line-decodable if whenever $\geq a$ scalars γ put $f_1 + \gamma f_2$ within δ of *some* codeword $U(\gamma)$, then $\geq b$ of those codewords lie on a single line $\{u_1 + \gamma u_2\}$ of codewords. **[PROVEN] (GG25 Thm. 3.5 = ABF Thm. 4.21):** if C is $(\delta, a, n+1)$ -line-decodable then $\varepsilon_{\text{mca}}(C, \delta) \leq a/|F|$. This is the entire positive route: a line-decoding theorem with threshold $a \leq 2^{-128}|F|$ at radius δ certifies $\delta_C^* \geq \delta$. The exact conjectural target is **CGHLL26 (eprint 2026/532, App. A.5, Conjecture 2):** RS over any prime-field domain is line-decodable up to the list-decoding capacity radius R_{cap} (CGHLL’s “Elias radius”, stated in the large- q -approximation convention — §1) with threshold $a = \ell(\theta)n + o(n)$, $\ell(\theta) \leq c_1 2^{c_2 H(\rho)/\eta}$ ($c_1 = c_2 = 1$ calibrated to KK25); “the same is true if the alphabet is from an extension field of F_p ” — originally stated without evidence, and now given its first concrete (negative-side) support: the KK25 worst case extends verbatim to genuine F_{p^ϵ} and sits *at/below* the threshold a below R_{cap} , exactly as on primes (§3.2, **n2-verdict.md**), so the extension clause is **consistent-and-tight**, not refuted.

3.1 Core-bet verdict

[Assessment, from line-decoding-analysis.md §7] The *strong* (up-to-MDS-capacity, small-error) core bet is **DEAD**: CS25 Cor. 1 gives $\varepsilon_{\text{ca}} = 1$ below $1 - \rho$, and Kambiré gives explicit smooth-prime bad lines at $(1 - \rho) - \Theta(1/\log n)$. The achievable ceiling is the **list-decoding capacity radius** R_{cap} , short of MDS capacity by $\sim 1/\log_2 q$. Moreover Thm 1.9 collapses the positive *small-error* MCA route into an equally hard list-decoding problem (§2.3), so the realistic positive form is **large-error** line-decodability up to $R_{\text{cap}} - \Theta(1/\log_2 |F|)$, with error $\varepsilon_{\text{mca}} = 2^{O(H(\rho)/\eta)} n/|F|$ exponential in $1/\eta$ (CGHLL §5.5: $\approx 0.83\beta - 0.91\beta$ bits/query, not the full β). The pessimistic scenario $\delta_C^* = J$ is **forced in char-2** (BCHKS Cor. 1.7: a bad line at $\delta = J$ with $a \geq n^{2-\epsilon}$, via subspace polynomials) but **unsupported in odd-characteristic prime fields** (no known construction puts a smooth-prime bad line in $(J, (1 - \rho) - O(1/\log n))$; the subspace-polynomial mechanism has no prime analogue — CGHLL’s central thesis). So the core bet, properly stated as *large-error line-decodability up to R_{cap}* , is **likely TRUE for smooth prime RS** but **currently unproven**, reducing to the smooth-domain list-size sub-lemma P’ below.

3.2 The extension-field case is settled at ESTABLISHED-MODULO-FORMALIZATION grade: the counterexample mechanism is field-agnostic

The smooth-domain negatives are stated over prime fields (Kambiré: Linnik primes $p \equiv 1 \pmod n$; BCHKS Thm 1.13: prime/Mersenne; KK25 Lemma 9: multiplicative subgroup of a *prime* field), and the char-2 results (BCHKS Cor.

1.7) are the opposite extreme. The intermediate case — RS over a smooth subgroup of an *odd-characteristic extension field* F_{p^e} (M31², BabyBear/Goldilocks extensions — the actually-deployed FRI fields) — was once flagged as the untouched opening. **It is now settled at EMF grade (n2-verdict.md, exact computation): the smooth-domain counterexample mechanism is FIELD-AGNOSTIC.** The firing condition ($e_2 = 0$), the bad scalar ($\lambda = e_1$), and the distinct-bad-scalar count are **characteristic-zero cyclotomic invariants** — e_1, e_2 are reductions mod the characteristic p of fixed elements of $\mathbb{Z}[\xi_s]$, independent of the extension degree e — so genuine odd-char extensions $GF(p^e)$ realize the *same* count as the prime field $GF(p)$, for primes outside the finite bad set excluded by the KK25 distinctness calibration (verified exactly, 10/10 unsaturated cases; a decoder-free certified bad line exists on genuine $GF(31^2)$, $GF(127^2)$ at $\delta = \frac{1}{2}, \frac{3}{4}$). The “no additive subspace” hope does not buy extension fields any extra safety, because the obstruction here is **multiplicative** (cyclotomic subset sums), not additive, and it lives in F_{p^e} identically to F_p .

Three consequences (all **ESTABLISHED-MODULO-FORMALIZATION**, fail-closed): (i) the negative extends to genuine odd-char extensions at the *same* constant- $\eta = 2/s$ band as primes — extensions are **not** a haven; (ii) the construction is **sub-threshold at 256-bit** for every target rate (the distinct-bad-scalar count needs $s \gtrsim 512$ to exceed 2^{128} under the rigorous binomial count — even the most optimistic characteristic-zero count needs $s \approx 162$ (§2.4) — but the Kambiré distinctness/Linnik window needs $s \lesssim 85$ at 256-bit — empty intersection; field-bit boundary $b^* \approx 140$, a witness existing only for $|F| \leq 2^{140}$ where the threshold degenerates), identical for prime and extension; (iii) it does **not** refute CGHLL Conj. 2’s extension clause — the firing radius $\delta = \text{capacity} - 2/s$ sits *below* R_{cap} (since $\eta = 2/s > 1/\log_2 p$, with margin far exceeding the ≤ 0.0017 large- q -approximation gap of §1), and the count $\approx 2^{H(2\rho)/(2\eta)}$ is at/below the conjecture’s exponential threshold $a = 2^{H(\rho)/\eta} n$, so the extension clause is *consistent-and-tight* exactly as the prime clause. **There is therefore no field-type opening: extensions are neither a haven nor a special vulnerability.** The genuine remaining frontier is the *positive* keystone P’ (§3.5). Note the asymmetry: the **negative endpoint mechanism is field-agnostic** (this section), but the positive **P’** route should first be proved for **prime-field multiplicative subgroups**, with the extension-field analogue stated separately (open) — P’ is *not* itself field-agnostic.

3.3 Sub-lemma N1 (extend Kambiré to $\rho = \frac{1}{2}$) — CONDITIONAL CLOSURE modulo a named distinctness extension (KK25 Lemma 9 beyond its stated range)

Sub-lemma N1 (n1-rho-half.md). *Over a smooth $\langle \omega \rangle \subset F_p$, $n = 2^t$, the line $f = X^{rm}, g = X^{(r-1)m}$ at rate exactly $\rho = \frac{1}{2}$ ($r = s/2 + 2$) has $\geq n^C$ proximates at $\delta = \frac{1}{2} - \eta$, $\eta = 2/s = \Theta(1/\log n)$, yet $\Delta([f, g], C^2) > \delta$. Verdict — conditional closure modulo a*

named distinctness extension (KK25 Lemma 9 beyond its stated range). The two **unconditional** halves go through *verbatim* at $\rho = \frac{1}{2}$: (i) the degree obstruction $\deg g = (r-1)m > k$, and (ii) the structural no-CA bound $S^* \leq (r-1)m < rm$ (both pure root-counting, rate- and field-blind). The bad-scalar count is supplied by the **PROVEN** closed form $|H^{(+r)}| = \sum_u \binom{s/2}{u} 2^u = 2^{\Theta(s)}$ (antipodal-pair / $\{-1, 0, +1\}$ -vector bijection, verified vs brute force at $s = 8, 16, 32$). Kambiré’s own count $\binom{s/2}{r}$ is **identically zero** at $\rho = \frac{1}{2}$ (since $r = s/2 + 2 > s/2$) and **KK25 Lemma 9 is out of range** ($r \leq \varphi(m)/2 = s/4$ is violated); the true count is *larger*, not absent. **Payoff (in Kambiré’s $\deg \leq k$ convention — under strict $\deg < k$ the construction fires not at exact rate $\frac{1}{2}$ but at $\frac{1}{2} + 1/n$, an $O(1/n)$ slack; see the residual below):** $\delta_C^*(\frac{1}{2}) \leq \frac{1}{2} - \Theta(1/\log n)$ — concretely ≤ 0.4375 at 128-bit; at 256-bit **not threshold-established** (R13: count $3^{32} = 2^{51} \ll 2^{128}$; 0.46875 mechanism-asymptotic reference only, power-of-two convention $s = 64$ — §2.4) — at the single most-deployed rate. **Field-agnostic corollary.** The count is a characteristic-zero cyclotomic invariant, so genuine odd-char extensions F_{p^e} give the *identical* count (25 distinct close scalars at the tested instance over $GF(p^2)$); the $\rho = \frac{1}{2}$ endpoint extends to the deployed extension case (**n2-verdict.md** mechanism). **The single residual (named).** A routine, numerically-airtight-but-not-yet-formalized strengthening of KK25 Lemma 9’s $\{-1, 0, +1\}$ -cyclotomic-independence / resultant-distinctness bound past its stated $r \leq \varphi(m)/2$ range to $r = s/2 + 2$ — the *for-all-n* distinctness-mod- p guarantee. The resultant criterion ($\text{Res}(\Phi_s, Q) = 0 \Leftrightarrow$ collision, $|\text{Res}| \leq (2r)^{s/2}$) is verified with **zero** counterexamples across $s \in \{8, 16, 32\}$, but Lemma 9 *as a black box* does not cover $\rho = \frac{1}{2}$. The result also uses Kambiré’s own $\deg \leq k$ convention (strict $\deg < k$ gives zero firing at rate exactly $\frac{1}{2}$; the construction lives at rate $\frac{1}{2} + 1/n$, an $O(1/n)$ slack vanishing asymptotically). **Honest prize framing.** This is a **near-capacity** endpoint, *not* a closure of the open band: it forecloses only $(0.4375, 0.5)$ at $\rho = \frac{1}{2}$ /128-bit (at 256-bit nothing is foreclosed at threshold — R13); it does **not** touch the open (J, R_{cap}) band ($= (0.293, 0.492)$ at 128-bit / $(0.293, 0.496)$ at 256-bit) and is **consistent with** (does not refute) CGHLL Conj. 2 ($\delta_{\text{unsafe}} < R_{\text{cap}}$, count at/below the threshold). It supplies a conditional negative endpoint at the most-deployed rate; the genuine frontier remains the positive keystone P’. **Never “unconditional” / “proven” tout court — it is CONDITIONAL on the named Lemma-9 extension.**

3.4 Sub-lemma N2 (extend Kambiré to an odd-characteristic extension field) — SETTLED (ESTABLISHED-MODULO-FORMALIZATION: field-agnostic, sub-threshold, Conj-2-consistent)

Sub-lemma N2. *Originally posed as “construct an analogous bad line over a smooth subgroup $\langle \omega \rangle$ of an odd-characteristic extension field F_{p^e} .” Settled by exact computation (`n2-verdict.md`) at EMF grade; no longer an open attack. **Close-out.** The counterexample mechanism is **field-agnostic**: the distinct-bad-scalar count is a characteristic-zero cyclotomic invariant (e_1, e_2 are reductions mod p of fixed elements of $\mathbb{Z}[\xi_s]$, independent of the extension degree e), so $GF(p^e)$ realizes the *same* count as $GF(p)$ (verified exactly, 10/10 unsaturated cases; the relevant obstruction is the *multiplicative* relation among the ζ^i , which behaves identically over F_p and F_{p^e}). The bad line **does extend** structurally to genuine extensions (decoder-free certificate at $\delta = \frac{1}{2}, \frac{3}{4}$ over $GF(31^2), GF(127^2)$), but it is **sub-threshold at 256-bit** (count needs $s \gtrsim 512$ to exceed 2^{128} , distinctness needs $s \lesssim 85$ — empty window; boundary $b^* \approx 140$) and it does **not** refute CGHLL Conj. 2’s extension clause ($\delta = \text{capacity} - 2/s < R_{\text{cap}}$, count $\approx 2^{H(2\rho)/(2\eta)}$ at/below the threshold $a = 2^{H(\rho)/\eta n}$). **Net: extensions are neither a haven nor a special vulnerability — N2 is settled at EMF grade, field-agnostic.***

3.5 The positive keystone P’ and the recommended priority

Sub-lemma P’ (smooth-domain small list just beyond Johnson — the positive keystone). *For $L = \langle \omega \rangle \subset F_p$, $|L| = n$, rate ρ , there is $\varepsilon_0 = \varepsilon_0(\rho) > 0$ with*

$$|\Lambda(\text{RS}[F_p, L, \rho n], J + \varepsilon_0)| \leq O_\rho(1/\varepsilon_0) \quad (\text{constant in } n).$$

Conclusion / payoff: via the GG25 Lemma-8 bootstrap (CGHLL A.5) this supplies line-decodability, hence $\varepsilon_{\text{mca}}(\text{RS}[F_p, L, \rho n], J + \varepsilon_0) \leq a(n)/|F| \leq 2^{-128}$ — the **first provable smooth-domain MCA point strictly beyond Johnson**. It is *weaker than CGHLL Conj. 1* (only ε_0 beyond J , not all the way to R_{cap}), dodging the hardest part of the barrier. **Field scope (not field-agnostic).** P’ as stated is a **prime-field multiplicative-subgroup** statement ($L = \langle \omega \rangle \subset F_p^\times$): the char-2 analogue is *false at Johnson* (BCHKS Cor. 1.7, subspace polynomials), and the genuine odd-char **extension-field** analogue is **OPEN** (extensions contain F_p -subspaces, so the “no additive subspaces” lever the prime route relies on is unavailable) — it must be stated separately. So, unlike the negative endpoint mechanism (which *is* field-agnostic, §3.2), P’ is *not* field-agnostic.

Recommended priority (line-decoding-analysis.md §7): (1) **P' first** — the positive keystone (a **prime-field multiplicative-subgroup** statement; *not* field-agnostic — its extension-field analogue is a separate, open problem), now the *single* genuine open attack: it simultaneously advances the list-decoding challenge and unlocks the MCA line-decoding theorem. (N1 carries a **conditional closure modulo the named KK25 Lemma-9 extension** (n1-rho-half.md) — the negative $\rho = \frac{1}{2}$ endpoint; N2 is **settled at EMF grade**, field-agnostic — neither is a remaining open attack.) A first target band for P': $\delta = J + \varepsilon_0$ for the smallest ε_0 beating Johnson, at $\rho = \frac{1}{2}$, M31 (the prime-field case); the genuine odd-char extension analogue is stated separately and remains open.

4. List-size companion records

The list-size bracket itself — the budget trichotomy, the Proposition with parts (A)/(B)/(C), and the Thm 1.9 separation — is the short paper's §3–§4. The ledger records the supporting tool-state items.

Johnson-floor constant provenance. The floor constant $1/(2\eta\rho)$ is ABF Cor. 3.3 exactly, verified in `listsize-analysis.md`; the ledger's VERIFY register still carries a stale `pre-listsize-analysis.md` row — “List size $n \cdot m/\eta$... VERIFY constant + exponents” — which is superseded by that verification.

Resolver state (R29/M2 follow-up). The resolver's $\delta^*[\text{Cap}]$ column (insensitive to $n \in \{2^{16}, 2^{20}, 2^{24}\}$ and $m \in \{2, 3\}$ to within the last digit; formerly computed under the alphabet cap `_QCAP_LOG2 = 200` that made its 256-bit entries 200-bit values — the cap is now 320, and the resolver plus `listsize-analysis.md` §6/§8 have been regenerated to the true $q = 2^{256}$ values) reports the *finite- n volume-bound crossing* $\approx R_{\text{cap}}$. That finite- n crossing is **rate-dependent in sign** relative to the large- q approximation $1 - H_q(\rho)$ at $n = 2^{20}$, 256-bit: $\approx 4.1 \cdot 10^{-7}$ *above* it at $\rho = \frac{1}{2}$, and $\approx 2 \cdot 10^{-5}$ *below* it at $\rho \leq \frac{1}{4}$ (verified by direct computation; the old “sits below” reading was wrong at $\rho = \frac{1}{2}$). This finite- n crossing is the **conjectural upper reach**, not a proven δ^* — the proven object is the bracket $[J - o(1), R_{\text{cap}}]$ (short paper §3). The regenerated 256-bit large- q -approximation values are 0.49609/0.74683/0.87288/0.93618 at $\rho = \frac{1}{2}/\frac{1}{4}/\frac{1}{8}/\frac{1}{16}$; the exact crossings R_{cap} are 0.49609/0.74681/0.87285/0.93616, differing at the fifth decimal for $\rho \leq \frac{1}{4}$. The 192-bit entries were unaffected by the old cap ($192 < 200$).

What remains open in sub-problem 2 [CONJECTURED]. A single open input governs the upper reach to $\approx R_{\text{cap}}$ for $|F| > 2^{128}$: can the interleaved list be proven $\leq$ a small constant (or $\text{poly}(n)$, i.e. $\leq B^{1/m}$) in the *worst case* for *smooth-domain* RS at radii between $J \approx 0.293$ and $R_{\text{cap}} \approx 0.492$ (128-bit, $\rho = \frac{1}{2}$)? (It does **not** bear on the knife-edge $|F| = 2^{128}$ *value*, which is already pinned at the UD supremum, short paper Prop. (B): at $B = 1$ any list of size 2 breaks the budget, and size-2 lists exist just past UD by the MDS midpoint

argument — a small-but- > 1 list past J , which is what the conjecture would deliver, cannot move it.) This is the smooth-domain interleaved-list conjecture (`listsize-analysis.md` §4), equivalent to derandomizing the random-RS list bound (ABF Thm. 3.6 / [AGL24]: $|\Lambda(C, 1 - \rho - \eta)| \leq 2(1 - \rho - \eta)/\eta$ w.h.p. over a *random* domain) to the structured smooth domain — ABF open problem §7.9 / our sub-lemma P'. It is itself a *list-decoding* statement (not a proximity-gap statement), so it too dodges Thm 1.9; it is gated instead by the different (also hard) open problem of RS list-decoding on structured domains. Note this is precisely the bound `line-decoding-analysis.md` §3 flags as a theorem only for random domains; it is **not** currently available for smooth domains. The experiment supports the conjecture's strong (constant-list) form at tiny fields (§5); no proof exists for cryptographic fields, and the Wave-1 experiments cannot see the cryptographic-scale deep-hole regime.

5. Experimental corroboration

Workstream C (`experiments/small_rs_atlas/`) is **exact full-codeword enumeration** over small fields ($q \leq 257$, $n \leq 256$) — *corroborative, not cryptographic-scale*. Three findings bear on the analysis above.

(a) **Smooth $\approx$ random/full everywhere [EXPERIMENTAL]**. Across 144 matched subgroup-vs-random cells, the smooth-random differential in mean close-count per line is $+0.037$ (std 0.207) and in bad-line fraction is -0.0014 (std 0.021) — centered on zero, bounded by sampling noise. Smooth subgroups are statistically indistinguishable from random domains; structured inputs (rational, degree- $k+1$) produce *fewer* bad lines than uniform-random, not more. This supports the positive / derandomization route (§3.1).

(b) **Onsets [EXPERIMENTAL]**. Bad lines (high close-count, no CA) onset essentially *exactly at Johnson* (mean onset — Johnson: subgroup -0.001 , random -0.005), identically across domain types. But **interleaved lists stay singleton far past Johnson**: the max interleaved list size is exactly 1 from below Johnson, through Johnson, departing from 1 only at $\approx$ capacity $-O(1/n)$ (the `interleaved_onset.csv` data give onset — capacity ≈ -0.05 to -0.10 , i.e. $c \approx 1-2$ in capacity $-c/n$) and exploding *at* capacity. $m = 3$ matches $m = 2$ exactly. This directly corroborates the sub-problem-2 benign-list story (short paper §3): the dangerous radius for interleaved lists is capacity $-O(1/n)$, not Johnson.

(c) **The Kambiré mechanism is a genuine smooth-specific algebraic excess — near capacity only [EXPERIMENTAL]**. Reproducing the monomial line $f = X^a, g = X^b$ on a smooth prime subgroup: the cleanest cell is $p = 257$, $n = 64$, $k = 2$, $\delta = 0.90625$ (= capacity -0.0625 , capacity = 0.9688): **SMOOTH close-count = 32 = predicted sumset size, vs matched RANDOM = 7** (counterexample_kambire.csv, smooth row at $\delta = 0.90625$ has `close_count=32`, `predicted_sumset_size=32`; random row

has `close_count=7`). This is a real smooth-specific excess corroborating the sunset mechanism — but **near capacity only**; below capacity $-O(1/\log n)$ no smooth-specific bad line appears.

Caveats [stated honestly]. (1) Field sizes are tiny ($q \leq 257$); the $1/|F|$ floor is huge, so 2^{-128} -scale effects are invisible — these are *structural* statements, not quantitative bounds at cryptographic scale. (2) Sampling over words/lines/targets (each distance/list exact); a rare structured configuration could be missed. (3) **Crucially, the tested “extension” domains in *this Wave-1 battery* were $\text{GF}(2^m) = \text{char-2}$** , which has its *own* known negatives (subspace polynomials) — **NOT odd-characteristic F_{p^e}** . The odd-char extension case (N2) is **not** probed by this Wave-1 battery; it is settled separately by the exact cyclotomic computation of `n2-verdict.md` (§3.2, EMF grade), which establishes the counterexample mechanism is **field-agnostic** — genuine F_{p^e} behaves exactly like F_p , sub-threshold at 256-bit, Conj-2-consistent. (4) The per-line common-agreement maximum S^* is an exact proxy for CA on a line, not the protocol’s ε_{mca} ; bridging the two is the theory step.

6. Status register

Claim	Status	Basis
$\delta_{\text{known-positive}} = J$ for smooth RS (all fields)	PROVEN	Bordage–Chiesa 2025/2051 Thm. 9.2; BCHKS 2025/2055 Thm. 1.5
Bordage–Chiesa MCA error $(m+\frac{1}{2})^7 n^2 d / (3\rho^{3/2} F)$	PROVEN	2025/2051 Thm. 9.2
BCHKS Thm 1.5 constant C_ρ (linear- n bound)	OUR-DERIVED (VERIFY)	η -exponent 5 firm; C_ρ a placeholder

Claim	Status	Basis
$\delta_{\text{known-negative (prime)}}$ $= (1 - \rho) - 6/\log_2 F $ — the headline continuum endpoint	INFERRED (finite-field inferred ceiling; VERIFY) — inferred arithmetic from the $8^s \leq F $ window, retained as asymptotic reference only; <i>not</i> a proven endpoint. The per-field threshold ceiling $(1 - \rho) - 2/s_{\max}(b)$ is the assembled Lemma (§2.2, R13) — assembled from cited components; KK25 cited not re-proved, $\rho = \frac{1}{2}$ N1-conditional PROVEN over the non-smooth domain F_q^* ($q - 1 = 2(2^{30} - 1)$ not a power of two); smooth-subgroup landing VERIFY — <i>not</i> a proven smooth-domain prize witness	Kambiré arXiv:2604.09724 Thm. 1 (asymptotic, proven) + our window arithmetic (inferred); per-field lemma assembled R13 (experiments/small_rs_atlas/n2_prize_threshold. the M31 $\delta = \frac{1}{2}$ point (next row) is the proven negative witness
M31, $\rho \approx \frac{1}{2}$ unconditional CA failure at $\delta = \frac{1}{2}$ (loss ≈ 0.016)	$\rho = \frac{1}{2}$ N1-conditional PROVEN over the non-smooth domain F_q^* ($q - 1 = 2(2^{30} - 1)$ not a power of two); smooth-subgroup landing VERIFY — <i>not</i> a proven smooth-domain prize witness	BCHKS Thm. 1.13 (M31 instantiation); caveat per <code>bchks.md</code> and §2.2
Thm 1.9 barrier: MCA-beyond-Johnson $\Rightarrow$ RS list-decoding- beyond-Johnson	PROVEN	BCHKS Thm. 1.9 = ABF Thm. 5.2

Claim	Status	Basis
$\delta_{\text{known-negative}}$ for genuine odd-char extension fields $= (1 - \rho) - 6/\log_2 F $ (same band as prime)	ESTABLISHED-MODULO-FORMALIZATION (field-agnostic mechanism) — the <i>mechanism</i> /count is field-agnostic and exact; the continuum <i>endpoint value</i> is the same finite-field inferred ceiling as the prime row (VERIFY for the continuum, asymptotic reference only; the per-field values follow the same assembled Lemma of §2.2, R13)	n2-verdict.md (Theorem + VERIFICATION): bad-scalar count is a char-0 cyclotomic invariant (exact); extends to $GF(p^e)$, sub-threshold at 256-bit, does not refute CGHLL Conj. 2
N1: $\rho = \frac{1}{2}$ endpoint $\delta_C^*(\frac{1}{2}) \leq \frac{1}{2} - \Theta(1/\log n)$ ($\leq 0.4375@128$; @256 not threshold-established per R13, 0.46875 asymptotic reference) — in Kambiré's $\deg \leq k$ convention (strict $\deg < k$: zero firing at exact $\rho = \frac{1}{2}$; construction lives at $\rho = \frac{1}{2} + 1/n$) — prime and F_{p^e}	CONDITIONALLY ESTABLISHED — conditional closure modulo a named distinctness extension (KK25 Lemma 9 beyond its stated range)	n1-rho-half.md : no-CA + degree obstruction verbatim/unconditional at $\rho = \frac{1}{2}$; count $ H^{(+r)} = \sum_u \binom{s/2}{u} 2^u = 2^{\Theta(s)}$ proven (replaces Kambiré's vacuous $\binom{s/2}{r}$), KK25 Lem. 9 out of range). Residual: extension of Lem. 9's $\{-1, 0, +1\}$ -independence/resultant bound past $r \leq \varphi(m)/2$ (zero counterexamples, $s \leq 32$). Near-capacity band only; consistent with CGHLL Conj. 2

Claim	Status	Basis
Per-field negative ceilings $\delta_C^* \leq (1 - \rho) - 2/s_{\max}(b)$, $s_{\max} = 16/16/32$ at $b = 31/64/128$ — at $\rho = \frac{1}{2}$: 0.375@31-bit (anchored at BabyBear; literal M31 has no smooth domain of order ≥ 4 — value transfers via the deployed circle/ F_{p^2} subgroup, i.e. 31-bit characteristic / 62-bit alphabet , at N2's ESTABLISHED- MODULO- FORMALIZATION grade, §2.2 Lemma caveat; M31 “almost closed” withdrawn ; window $[0.293, 0.375]$, radius-level — sub-256-bit fields carry no 2^{-128} positive certificate, §2.1/§2.4), 0.375@64-bit, 0.4375@128-bit 256-bit negative ceiling below R_{cap} (Kambiré-type near-capacity, constant- η ; any known mechanism) at $\varepsilon^* = 2^{-128}$; feasibility boundary $b^* \approx 140$ bits	ASSEMBLED (KK25 cited not re-proved; $\rho = \frac{1}{2}$ N1-conditional; deg-convention tracked; the literal-M31 instantiation N2-grade per the §2.2 caveat; radius-level brackets per §2.4)	assembled Lemma §2.2 (R13): Kambiré Thm. 1 count + KK25 distinctness + experiments/small_rs_atlas/n2_prize_threshold. problem-ledger.md §5–§6
256-bit negative ceiling below R_{cap} (Kambiré-type near-capacity, constant- η ; any known mechanism) at $\varepsilon^* = 2^{-128}$; feasibility boundary $b^* \approx 140$ bits	OPEN — no threshold-established ceiling below R_{cap} (the generic CS25/Elias ceiling at $\approx R_{\text{cap}}$, §2.2, stands at every field size); rescue paths assessed: per-prime certification blocked ($\sim 2^{77}$ enumeration; norm bound $2^{676} > 2^{256}$), BGK-type growth open (≈ 113 -bit gap)	ledger VERIFY register (2026-06-10); literature/notes/subgroup-sumset-growth.md; experiments/small_rs_atlas/results/n2_prize_th

Claim	Status	Basis
List-size trichotomy; crossover at $ F = 2^{128}$	OUR-DERIVED	arithmetic on B ; Johnson + Elias composition (short paper §3)
$\delta_C^{*(2)} = (1 - \rho)/2$ at $ F = 2^{128}$ (binding/UD) — as a supremum ; with closed balls the endpoint is not attained (discreteness)	PROVEN	short paper Prop. (B); MDS/Singleton
$\delta_C^{*(2)} \geq J - o(1)$ for $ F > 2^{128}$ (proven floor; precisely $J - \eta_{\min}$, $\eta_{\min} = 1/(2\rho B^{1/m})$)	PROVEN	Johnson Cor. 3.3 + interleaving; smooth domains
$\delta_C^{*(2)} \leq R_{\text{cap}}(q, \rho) =$ $H_q^{-1}(1 - \rho)$ for $ F > 2^{128}$ (proven ceiling — the inverse-entropy crossing; the large- q approximation $1 - H_q(\rho)$ differs from it beyond the fifth decimal at some rates, §1) $\Rightarrow$ proven bracket	PROVEN	Elias/CS25 lower-bound ceiling (list $\geq q^{\Omega(n)} \gg B$ above the crossing)
$\delta_C^{*(2)} \in [J - o(1), R_{\text{cap}}]$ for $ F > 2^{128}$	PROVEN BRACKET	compose the two rows above; short paper §3
Upper reach $\delta_C^{*(2)} \approx R_{\text{cap}}$ for $ F > 2^{128}$	CONJECTURAL	needs worst-case large-list (list $\leq B^{1/m}$) smooth-RS list-decoding beyond Johnson; ABF §7.9 / P'; not gated by Thm 1.9 but open
Sub-problem 2 decouples from Thm 1.9, with a standard list-bound bracket (vs sub-problem 1's barrier-gated state)	OUR-DERIVED (rigorous)	short paper §4 (it is a pure list statement)
Smooth-domain interleaved list $\leq P(\rho, \eta)^m$ up to capacity $-\eta$	CONJECTURED	needs structured-domain list-decoding (ABF §7.9)

Claim	Status	Basis
CGHLL Conj. 2: prime/extension RS line-decodable to R_{cap}	CONJECTURED (extension clause now <i>consistent-and-tight</i> , not refuted)	CGHLL 2026/532 App. A.5; <code>n2-verdict.md</code> : the KK25 counterexample extends to genuine F_{p^e} and sits at/below the threshold a below R_{cap} — first concrete (negative-side) support for the extension clause
Smooth $\approx$ random; onset at Johnson; interleaved singleton to capacity $-O(1/n)$; $m=3 \equiv m=2$	EXPERIMENTAL	small fields $q \leq 257$, corroborative
$p = 257$ smooth close-count 32 vs random 7 at $\delta = 0.906$	EXPERIMENTAL	<code>counterexample_kambire.csv</code> ; near-capacity only

7. What this ledger does and does not establish

- **The companion short paper establishes** (rigorous, from cited Johnson + Elias bounds) the **proven two-sided bracket** $\delta_C^{*(2)} \in [J - o(1), R_{\text{cap}}]$ **for the interleaved list-size grand challenge at** $|F| > 2^{128}$ (proven floor $J - \eta_{\min}$ with $\eta_{\min} = 1/(2\rho B^{1/m}) \leq 2^{-29}$ deployed, written $J - o(1)$; proven ceiling the inverse-entropy crossing R_{cap} ; e.g. $[0.293, 0.496]$ at $\rho = \frac{1}{2}$, 256-bit), the collapse to the UD supremum $(1 - \rho)/2$ at $|F| = 2^{128}$ (endpoint not attained — discreteness) and vacuity below — and, most importantly, that this bracket **uses no new list-decoding and therefore dodges the Thm 1.9 barrier**, demonstrating that the two grand challenges are not equivalent and that sub-problem 2 is decoupled from Thm. 1.9 with a standard list-bound bracket, while sub-problem 1 stays barrier-gated. The upper reach to $\approx R_{\text{cap}}$ is **conjectural** (it requires an unproven worst-case large-list smooth-RS list-decoding theorem beyond Johnson; ABF §7.9 / P'), not established anywhere in this work.
- **This ledger does record (ASSEMBLED, with named soft spots)** the assembled per-field MCA negative-ceiling **assessment** $\delta_C^* \leq (1 - \rho) - 2/s_{\max}(b)$ — at $\rho = \frac{1}{2}$: 0.375 at 31- and 64-bit, 0.4375 at 128-bit (§2.2 Lemma, R13; KK25 cited not re-proved, $\rho = \frac{1}{2}$ N1-conditional) — including the **withdrawal of the M31 “almost closed” reading** (honest window $[0.293, 0.375]$, not $[0.293, 0.306]$ — a radius-level window, §2.4:

sub-256-bit fields carry no 2^{-128} positive certificate; M31 row read as 31-bit characteristic / 62-bit alphabet, §2.2); and it records that at **256-bit no threshold-established negative ceiling below R_{cap} — i.e., no Kambiré-type near-capacity, constant- η ceiling — exists from any known mechanism** (the generic CS25/Elias ceiling at $\approx R_{\text{cap}}$ stands at every field size), with both rescue paths assessed (per-prime certification blocked at $\sim 2^{77}$ enumeration; BGK-type growth ≈ 113 bits short) and the feasibility boundary $b^* \approx 140$ bits (§2.4).

- **Neither artifact claims** any solution to the **MCA grand challenge**. For sub-problem 1, only the *bracket* $[J, R_{\text{cap}})$ is known (bounded above near capacity by Kambiré/CS25/BCHKS, the negative mechanism established field-agnostic at EMF grade — the near-capacity negative holds for genuine odd-char extensions at the same band as prime, `n2-verdict.md`); a positive MCA point strictly above J for smooth RS is unproven and is exactly the prize.
- **What remains for the full prize:** (1) **MCA, positive (the genuine frontier):** prove sub-lemma P' (small smooth-domain list just past J) $\Rightarrow$ via GG25 bootstrap, the first $\delta_C^* > J$; the realistic ceiling is large-error line-decodability to $R_{\text{cap}} - \Theta(1/\log_2 |F|)$ (CGHLL Conj. 2), unproven for structured domains. **P' should first be proved for prime-field multiplicative subgroups; its extension-field analogue is stated separately and is open — P' is *not* field-agnostic** (only the negative endpoint mechanism is). (2) **MCA, negative:** the $\rho = \frac{1}{2}$ endpoint now carries a **conditional closure modulo a named distinctness extension (KK25 Lemma 9 beyond its stated range)** — sub-lemma N1 ($\delta_C^*(\frac{1}{2}) \leq \frac{1}{2} - \Theta(1/\log n)$, prime and F_{p^e} , **in Kambiré's $\deg \leq k$ convention:** under strict $\deg < k$ the construction fires at rate $\frac{1}{2} + 1/n$, not exact $\frac{1}{2}$ — §3.3); it forecloses only the near-capacity band, not the open (J, R_{cap}) . (N2, the odd-char extension negative, is **settled at EMF grade:** field-agnostic, sub-threshold at 256-bit, Conj-2-consistent — `n2-verdict.md`; no longer an open attack.) The genuinely open negative item is the **256-bit ceiling below R_{cap}** (Kambiré-type near-capacity, constant- η): no known mechanism reaches the 2^{-128} threshold below R_{cap} at $b = 256$ (the generic CS25/Elias ceiling at $\approx R_{\text{cap}}$ stands) — per-prime certification blocked at $\sim 2^{77}$ enumeration, BGK-type growth ≈ 113 bits short (§2.4) — with feasibility boundary $b^* \approx 140$ bits. (3) **List-size, upper reach:** the smooth-domain interleaved-list conjecture (CONJECTURED; ABF §7.9) — it governs only the conjectural reach to $\approx R_{\text{cap}}$ for $|F| > 2^{128}$; the knife-edge $|F| = 2^{128}$ *value* is pinned at the UD supremum (short paper Prop. (B)) and is not open.

Structural work toward the positive side is ongoing.

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- [ABF] G. Arnon, D. Boneh, G. Fenzi, *Open Problems in List Decoding and Correlated Agreement*, IACR eprint **2026/680** (Apr 8 2026). The survey framing the Proximity Prize; Defs. 2.9, 2.12, 4.1, 4.3; Cor. 3.3; Lemma 6.6; Thms. 4.12, 4.17, 4.21, 5.2, 5.3; open problem §7.9.
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- [Kambiré] A. Kambiré, *Proximity Gaps Conjecture Fails Near Capacity over Prime Fields*, **arXiv:2604.09724** (Apr 2026). Thm. 1 (explicit smooth-prime counterexample at $(1 - \rho) - \Theta(1/\log n)$); the $f = X^{rm}$, $g = X^{(r-1)m}$ line; flesh-out of the Krachun–Kazanin (KK25) sketch.
- [CGHLL26] D. Carmon, L. Goldberg, U. Haböck, L. Lerer, I. Lesokhin (+ S. Papini, S. Samocha), *S-two Whitepaper*, IACR eprint **2026/532** (2026). App. A.5: the Elias radius (large- q approximation of R_{cap}); Conjecture 1 (list-decodability to it); **Conjecture 2** (line-decodability to it, the core bet, incl. extension-field clause); Thm. 37 (= KK25); Lemma 9 (distinct-sums count).

- **[Short paper]** Ö. Demirel, *Two Grand Challenges Are Not One*, `technical-note-short.md` (this repository, 2026-06-10). The polished companion: budget trichotomy, proven list-size bracket, Thm 1.9 separation, field-agnosticity of the negative mechanism, open problems.
- **Project artifacts (this work):** `problem-ledger.md` (formal bracket ledger); `listsize-analysis.md` + `calculator/listsize_resolution.py` (sub-problem-2 bracket analysis); `line-decoding-analysis.md` (core-bet verdict, sub-lemmas); `n1-rho-half.md` (sub-lemma N1 close-out: the $\rho = \frac{1}{2}$ negative endpoint, CONDITIONAL-CLOSURE modulo the named KK25 Lemma-9 extension, with the exact closed-form count); `n2-verdict.md` (sub-lemma N2 close-out: the field-agnostic extension case); `calculator/out/delta_star_tables.md` (machine-generated δ_C^* brackets; δ_{unsafe} column pre-R13, regeneration pending); `experiments/small_rs_atlas/` (exact enumeration, `counterexample_kambire.csv`, `interleaved_onset.csv`, `verify_rho_half_sums.py`); `experiments/small_rs_atlas/results/n2_prize_threshold.json` (the $b^* \approx 140$ feasibility boundary and per-field threshold arithmetic); `literature/notes/subgroup-sumset-growth.md` (the 256-bit rescue-path survey: BGK-type subgroup-sumset growth, ≈ 113 -bit gap).